```
    context.stroke();
}
```

Starting and pausing the tracking process

To start the tracking process, *tracking.js* provides a call to *start( )* and *stop( )* methods.

```
mT.stop(); //to stop tracking
mT.start(); //to start tracking
```

Setting up custom colours for tracking

As seen, the input to a colour tracker is a list of probable colours (e.g., [*yellow*]). As the definition suggests, a colour tracker must be able to track colours. *Tracking.js* provides a method *registerColor* that handles user-specified custom colours.

```
tracking.ColorTracker.registerColor('<color_name>' ,
callback_color);
```

The =*callback_color* callback will have input arguments as red, blue and green values. Since this is a custom colour, one has to define the RGB ranges. If the RGB argument meets the range, the function returns true, else it'll return false.

```
function callback_color(r , g, b){
    if(r > r_low && r < r_high && g > g_low && g < g_high &&
b > b_low && b < b_high){
      return true;
    }
    return false;
}
```

Here, *r_low, r_high*, etc, refer to the lower and upper bounds of the threshold values, respectively. Having registered the colour, one can simply append *color_name* to *color_list* in *tracking.ColorTracker (color_list)*.

Face tagging using *tracking.js*

Facebook has this feature whereby one can tag one's friends. There are different sets of mathematical frameworks developed to perform visual recognition as well as detection, of which one of the most robust options is the Viola-Jones Detection framework.

A brief introduction to Viola Jones: Each human face has multiple features, with many significant discernible visual differences which are the inputs that help in face recognition. These are known as Haar Cascades (which you can look up in */src/detection/training/haar*). Examples for significant variations in facial features include:

- Location of the eyes and nose
- Size of the eyes, nose, etc
- Mutual contrast between facial features

By training over such features, the detection framework

Figure 7: Face detection, tagging and tracking

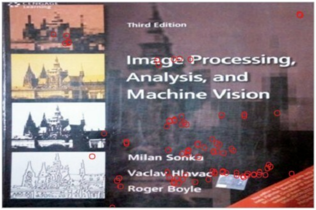

Figure 8: Feature points

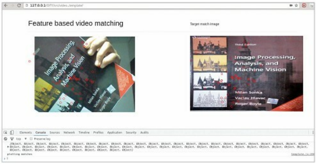

Figure 9: Matching via features

is made to locate areas of an image containing regions that satisfy the above constraints, thereby aiding in face detection.

To integrate your front-end with face recognition, *tracking.js* provides another script located in *build/data/face-min.js.* This basically loads the Viola Jones parameters over trained data, including *face-min.js* as well as *tracking.min.js* files.

To initialise and use the face tracker, type:

```
var face_tracker = new tracking.ObjectTracker("face");
face_tracker.setInitialScale(param_for_block_scaling);
face_tracker.setEdgesDensity(classifier_param_for_edges_
inside_block);
face_tracker.setStepSize(block_step_size);
var mTracker = tracking.track("#myVideo", face_tracker,
```